1. Write a query to calculate the total salary of all employees.

-> select sum(salary) as total_salary from employee;

```
mysql> select sum(salary) as total_salary from employee;
+--------------+
| total_salary |
+--------------+
|    375000.00 |
+--------------+
1 row in set (0.07 sec)
```

2. Write a query to find the average salary of employees in each department using GROUP BY.

-> select dep_id,avg(salary) as average_salary from employee group by dep_id;

```
mysql> select dep_id,avg(salary) as average_salary from employee group by dep_id;
+--------+----------------+
| dep_id | average_salary |
+--------+----------------+
|     10 |   72500.000000 |
|     20 |   57500.000000 |
|     30 |   57500.000000 |
+--------+----------------+
3 rows in set (0.07 sec)
```

3. Write a query to count the total number of employees in each department

-> select dep_id,count(*) as total_employee from employee group by dep_id;

```
ysql> select dep_id,count(*) as total_employee from employee group by dep_id;
--------+----------------+
 dep_id | total_employee |
--------+----------------+
     10 |              3 |
     20 |              3 |
     30 |              2 |
--------+----------------+
 rows in set (0.00 sec)
```

4. Write a query to display departments having more than 5 employees using HAVING clause.

->select dep_id,count(*) as total_employee from employee group by dep_id having total_employee > 5;

```
mysql> select dep_id,count(*) as total_employee from employee group by dep_id having total_employee > 5;
Empty set (0.01 sec)

mysql> select dep_id,count(*) as total_employee from employee group by dep_id having total_employee > 2;
+--------+----------------+
| dep_id | total_employee |
+--------+----------------+
|     10 |              3 |
|     20 |              3 |
+--------+----------------+
2 rows in set (0.00 sec)
```

5. Write a query to list distinct department locations from the Dept table.

-> select distinct location from department;

```
mysql> select distinct location from department;
+----------+
| location |
+----------+
| pune     |
| mumbai   |
+----------+
2 rows in set (0.06 sec)
```

6. Write a query to display the highest salary among all employees. Write a query to display all total number of employee in table.

-> select max(salary) as maximum_salary, count(*) as total_number_of_employee from employee;

```
mysql> select max(salary) as maximum_salary, count(*) as total_number_of_employee from employee;
+---------------+--------------------------+
| maximum_salary | total_number_of_employee |
+---------------+--------------------------+
|      85000.00 |                        8 |
+---------------+--------------------------+
1 row in set (0.00 sec)
```

7. Write a query to find employees whose name starts with 'A' using LIKE operator.

-> select * from employee where emp_name like 'A%';

```
mysql> select * from employee where emp_name like 'A%';
+--------+----------+----------+--------+--------------+
| emp_id | emp_name | salary   | dep_id | joining_date |
+--------+----------+----------+--------+--------------+
|    102 | Aniket   | 60000.00 |     10 | 2026-10-25   |
|    105 | Aditi    | 65000.00 |     20 | 2026-01-20   |
+--------+----------+----------+--------+--------------+
2 rows in set (0.07 sec)
```

8. Write a query to find employees whose name ends with 'n' using LIKE operator.

-> select * from employee where emp_name like '%n';

```
mysql> select * from employee where emp_name like '%n';
+--------+----------+----------+--------+--------------+
| emp_id | emp_name | salary   | dep_id | joining_date |
+--------+----------+----------+--------+--------------+
|    104 | navin    | 70000.00 |     30 | 2026-02-17   |
|    108 | nitin    |     NULL |     20 | 2026-03-15   |
+--------+----------+----------+--------+--------------+
2 rows in set (0.00 sec)
```

9. Write a query to find employees whose name contains 'ra' using LIKE operator.

->select * from employee where emp_name like '%ra%';

```
mysql> select * from employee where emp_name like '%ra%';
+--------+----------+----------+--------+--------------+
| emp_id | emp_name | salary   | dep_id | joining_date |
+--------+----------+----------+--------+--------------+
|    103 | Pranav   | 50000.00 |     20 | 2026-07-10   |
|    106 | Shravni  |     NULL |     10 | 2026-10-20   |
+--------+----------+----------+--------+--------------+
2 rows in set (0.00 sec)
```

10. Write a query to display all employees sorted by their Salary in descending order.

-> select * from employee order by salary desc;

```
mysql> select * from employee order by salary desc;
+--------+----------+----------+--------+--------------+
| emp_id | emp_name | salary   | dep_id | joining_date |
+--------+----------+----------+--------+--------------+
|    101 | vishva   | 85000.00 |     10 | 2026-08-20   |
|    104 | navin    | 70000.00 |     30 | 2026-02-17   |
|    105 | Aditi    | 65000.00 |     20 | 2026-01-20   |
|    102 | Aniket   | 60000.00 |     10 | 2026-10-25   |
|    103 | Pranav   | 50000.00 |     20 | 2026-07-10   |
|    107 | Mansi    | 45000.00 |     30 | 2026-09-26   |
|    106 | Shravni  |     NULL |     10 | 2026-10-20   |
|    108 | nitin    |     NULL |     20 | 2026-03-15   |
+--------+----------+----------+--------+--------------+
8 rows in set (0.00 sec)
```

11. Write a query to display all employees sorted by DeptID ascending and then Salary descending.

-> select * from employee order by dep_id asc, salary desc ;

```
mysql> select * from employee order by dep_id asc, salary desc ;
+--------+----------+----------+--------+--------------+
| emp_id | emp_name | salary   | dep_id | joining_date |
+--------+----------+----------+--------+--------------+
|    101 | vishva   | 85000.00 |     10 | 2026-08-20   |
|    102 | Aniket   | 60000.00 |     10 | 2026-10-25   |
|    106 | Shravni  |     NULL |     10 | 2026-10-20   |
|    105 | Aditi    | 65000.00 |     20 | 2026-01-20   |
|    103 | Pranav   | 50000.00 |     20 | 2026-07-10   |
|    108 | nitin    |     NULL |     20 | 2026-03-15   |
|    104 | navin    | 70000.00 |     30 | 2026-02-17   |
|    107 | Mansi    | 45000.00 |     30 | 2026-09-26   |
+--------+----------+----------+--------+--------------+
8 rows in set (0.00 sec)
```

12. Write a query to find employees whose salary is between 30,000 and 60,000.

-> select * from employee where salary between 30000 and 60000;

```
mysql> select * from employee where salary between 30000 and 60000;
+--------+----------+----------+--------+--------------+
| emp_id | emp_name | salary   | dep_id | joining_date |
+--------+----------+----------+--------+--------------+
|    102 | Aniket   | 60000.00 |     10 | 2026-10-25   |
|    103 | Pranav   | 50000.00 |     20 | 2026-07-10   |
|    107 | Mansi    | 45000.00 |     30 | 2026-09-26   |
+--------+----------+----------+--------+--------------+
3 rows in set (0.07 sec)
```

13. Write a query to display all employees whose DeptID is in (10, 20, 30).

-> select * from employee where dep_id in(10, 20, 30);

```
mysql> select * from employee where dep_id in(10, 20, 30);
+--------+----------+----------+--------+--------------+
| emp_id | emp_name | salary   | dep_id | joining_date |
+--------+----------+----------+--------+--------------+
|    101 | vishva   | 85000.00 |     10 | 2026-08-20   |
|    102 | Aniket   | 60000.00 |     10 | 2026-10-25   |
|    103 | Pranav   | 50000.00 |     20 | 2026-07-10   |
|    104 | navin    | 70000.00 |     30 | 2026-02-17   |
|    105 | Aditi    | 65000.00 |     20 | 2026-01-20   |
|    106 | Shravni  |     NULL |     10 | 2026-10-20   |
|    107 | Mansi    | 45000.00 |     30 | 2026-09-26   |
|    108 | nitin    |     NULL |     20 | 2026-03-15   |
+--------+----------+----------+--------+--------------+
8 rows in set (0.00 sec)
```

14. Write a query to display Min salary of employee

-> select min(salary) as minimum_salary from employee;

```
mysql> select min(salary) as minimum_salary from employee;
+----------------+
| minimum_salary |
+----------------+
|       45000.00 |
+----------------+
1 row in set (0.00 sec)
```

15. Write a query to display employees whose JoiningDate is between '2020-01-01' and '2021-12-31'.

-> select * from employee where joining_date between '2020-01-01' and '2021-12-31';

```
mysql> select * from employee where joining_date between '2020-01-01' and '2021-12-31';
+--------+----------+----------+--------+--------------+
| emp_id | emp_name | salary   | dep_id | joining_date |
+--------+----------+----------+--------+--------------+
|    109 | Omkar    | 70000.00 |     30 | 2021-02-15   |
+--------+----------+----------+--------+--------------+
1 row in set (0.00 sec)
```

16. Write a query to display employees whose Salary is NULL.

->select * from employee where salary is null;

```
mysql> select * from employee where salary is null;
+--------+----------+--------+--------+--------------+
| emp_id | emp_name | salary | dep_id | joining_date |
+--------+----------+--------+--------+--------------+
|    106 | Shravni  |   NULL |     10 | 2026-10-20   |
|    108 | nitin    |   NULL |     20 | 2026-03-15   |
+--------+----------+--------+--------+--------------+
2 rows in set (0.00 sec)
```

17. Write a query to display employees whose Salary is NOT NULL.

-> select * from employee where salary is not null;

```
mysql> select * from employee where salary is not null;
+--------+----------+----------+--------+--------------+
| emp_id | emp_name | salary   | dep_id | joining_date |
+--------+----------+----------+--------+--------------+
|    101 | vishva   | 85000.00 |     10 | 2026-08-20   |
|    102 | Aniket   | 60000.00 |     10 | 2026-10-25   |
|    103 | Pranav   | 50000.00 |     20 | 2026-07-10   |
|    104 | navin    | 70000.00 |     30 | 2026-02-17   |
|    105 | Aditi    | 65000.00 |     20 | 2026-01-20   |
|    107 | Mansi    | 45000.00 |     30 | 2026-09-26   |
|    109 | Omkar    | 70000.00 |     30 | 2021-02-15   |
+--------+----------+----------+--------+--------------+
7 rows in set (0.00 sec)
```

18. Write a query to calculate the total salary per department, but only for departments where total salary is greater than 1,00,000 (use HAVING).

-> select dep_id, sum(salary) as total_salary from employee group by dep_id having sum(salary) > 100000;

```
mysql> select dep_id, sum(salary) as total_salary from employee group by dep_id having sum(salary) > 100000;
+--------+--------------+
| dep_id | total_salary |
+--------+--------------+
|     10 |    145000.00 |
|     20 |    115000.00 |
|     30 |    185000.00 |
+--------+--------------+
3 rows in set (0.00 sec)
```

19. Write a query to display all distinct employee names.

-> select distinct emp_name from employee;

```
mysql> select distinct emp_name from employee;
+----------+
| emp_name |
+----------+
| vishva   |
| Aniket   |
| Pranav   |
| navin    |
| Aditi    |
| Shravni  |
| Mansi    |
| nitin    |
| Omkar    |
+----------+
9 rows in set (0.00 sec)
```

20. Write a query to count the number of departments having the same location.

-> select location, count(*) as total_department from department group by location;

```
mysql> select location, count(*) as total_department from department group by location;
+----------+------------------+
| location | total_department |
+----------+------------------+
| pune     |                2 |
| mumbai   |                1 |
+----------+------------------+
2 rows in set (0.07 sec)
```